

Review article

Conservative Functional Treatment of Acute Fibular Ligament Rupture of the Ankle

Hans Zwipp

Summary

Background: Acute rupture of the fibular ligament complex is one of the commonest injuries in sports. Prospective randomized trials in the 1980s led to a paradigm shift from primary surgical repair to conservative functional treatment.

Methods: This review is based on publications retrieved by a selective search in PubMed, Embase, and the Cochrane Library for randomized controlled trials (RCTs) and meta-analyses on surgical versus conservative treatment from the years 1983 to 2023.

Results: 10 of 11 prospective randomized trials of surgical versus conservative treatment conducted between 1984 and 2017 did not reveal any significant difference in the overall outcome. These findings were confirmed in two meta-analyses and two systematic reviews that were published between 2007 and 2019. Isolated benefits in the surgical group were outweighed by a variety of postoperative complications. The anterior fibulotalar ligament (AFTL) was ruptured in 58% to 100% of cases, followed by the fibulocalcaneal ligament combined with the LFTA in 58% to 85%, and the posterior fibulotalar ligament (mostly incomplete ruptures) in 1.9% to 3%.

Conclusion: Conservative functional treatment is now the standard treatment in acute fibular ligament rupture of the ankle because it is low-risk, low-cost, and safe. Primary surgery is indicated in only 0.5% to 4% of cases. Physical examination for tenderness to palpation and for stability, as well as stress ultrasonography, can be used to differentiate sprains from ligamentous tears. MRI is superior only for the detection of additional injuries. Stable sprains can be successfully treated with an elastic ankle support for a few days, and unstable ligamentous ruptures with an orthosis for 5 to 6 weeks. Subsequent physiotherapy with proprioceptive exercises is the best way to prevent recurrent injury

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Rupture of the fibular ligament complex of the ankle is one of the commonest injuries in sports, with 94% of the injured being under 40 years of age and 59% of them male (1–5). When the acquisition of evidence-based data was introduced to trauma surgery, prospective randomized trials involving acute fibular ligament tears in the early 1980s (6–11) led to a paradigm shift from primary surgical repair to conservative functional treatment. This spares the patient the risks of anesthesia and surgery yet achieves comparable ligament healing outcomes while, at the same time, saving enormous costs for the health care system (12). The distinction between a stable fibular ligament injury (stretched ligament, sprain) from an unstable ligament tear is crucial for the subsequent treatment, so more subtle clinical tests and diagnostic imaging are required. Given that unrecognized ligament tears can ultimately result in chronic instability (which has been wrongly attributed to conservative functional treatment), the aim of the present article was to examine whether more recent randomized controlled trials (RCTs) of surgical versus conservative management (13–17), meta-analyses (18, 19), and reviews (20, 21) differ from earlier RCTs (6–11).

Epidemiology, causes of injury, biomechanics

From an epidemiological perspective, an acute tear of the fibular ligaments is one of the commonest injuries in sports (1–3). In 2002 in the United Kingdom, with its 60 million people at that time, 300 000 sprains were recorded in accident and emergency units and 42 000 cases were classified as having clinically relevant instability (1).

Sports is reported as a cause of fibular ligament tears, especially those types with player-to-player contact, such as football, basketball and volleyball (1–3). The fibular ligaments can be stretched and, with progressive ankle tilt, torn due to excessive inward tilting of the foot (supination, adduction, inversion) with delayed counter-reaction of the foot evertors as a result of direct bodily contact in 42.4% associated with a misstep (3) or unexpected landing with the foot on that of another player.

According to Swenson et al. (3), 5373 ankle sprains were diagnosed in US American high-school athletes during 17 172 376 competitive exposures between 2005 and 2010, despite wearing ankle braces

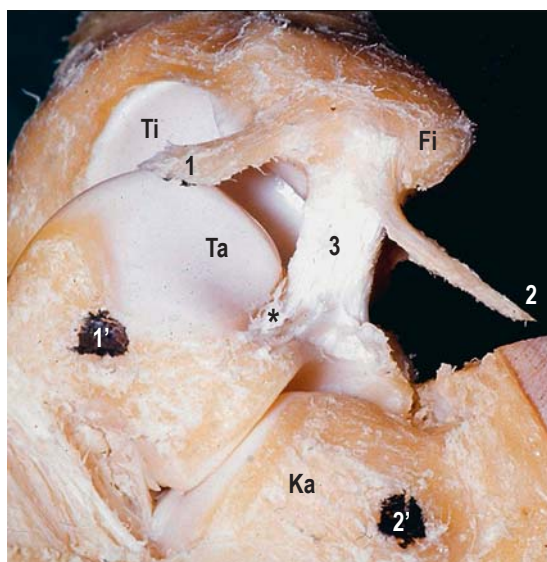


Figure 1: The three lateral ankle ligaments point in the three directions of space, as shown here on a left foot fixed in formalin after removal of the skin, subdermis, and peroneal tendons. The anterior talofibular ligament (1) has been sharply detached at its insertion (1', marked black here) on the talus (Ta), immediately anterior to the cartilaginous lateral malleolar facet of the talus. The medial calcaneofibular ligament (2) has also been sharply detached at its insertion (2', marked black here) on the calcaneus (Ka), so that when the talocrural joint is opened up, the posterior talofibular ligament (3) is seen as an extremely strong ligament, which, as the hindfoot is progressively forced into eversion and external rotation, is gradually torn at its short fibers (*) along Fawcett's line on the posterior margin of the talus (partial tear) and only disrupts completely in extremely rare cases. The talus is seen as it laterally falls out of the ankle joint between tibial plafond (Ti) and the fibula (Fi)

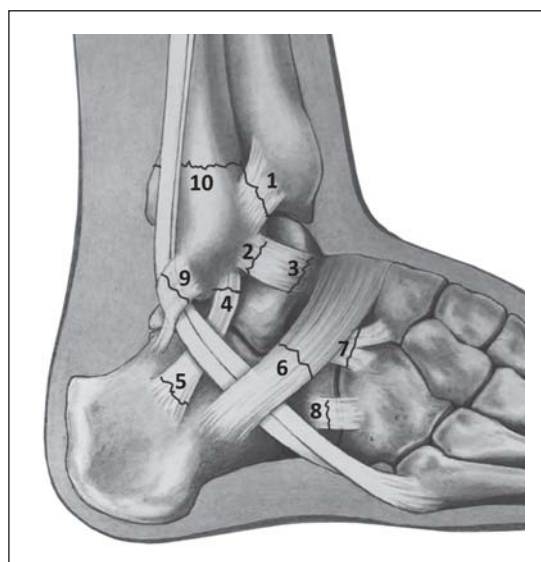


Figure 2: Diagnostic palpation of the tender points after acute injury: 1. anterior syndesmotic ligament; 2-3. anterior talofibular ligament; 4-5. calcaneofibular ligament; 6. distal peroneal retinaculum and tarsal sinus; 7. bifurcate ligament; 8. lateral calcaneocuboid ligament; 9. proximal peroneal retinaculum; 10. distal epiphyseal plate in the child and adolescent

in 10.6% of the cases, corresponding to an incidence of 3.13 per 10 000 exposures. The likelihood of an ankle sprain was calculated with a median injury risk ratio (RR) of 5.88 [95% confidence interval: 5.34; 6.48] for American football, of 3.55 for soccer, and 2.22 for basketball.

Between 2012 and 2021, the German Federal Statistics Office (Destatis) registered on average 2989 (2603–3203) patients per year with an ICD-10 diagnosis of S93.2 (fibular ligament tear). This reflects the number of inpatients, i.e. those operated (4). Because Destatis, accident insurance companies, and health insurance funds had no data on outpatients with an ICD-10 diagnosis of S93.2, i.e. those treated conservatively, a doctoral thesis (5) was launched in 2014 in the form of a nationwide online study which identified 51 503 patients treated conservatively for a fibular ligament tear and 2997 patients who had been operated on, which correlated well with the 3002 patients reported by Destatis (4) for the year 2014, despite the high risk of bias inherent in this type of study.

Three fibular ligaments provide passive biomechanical stability to the lateral ankle in the three

directions of space (Figure 1). The peroneal tendons provide active stability to the lateral joint, which is why an intact proprioceptive reflex arc is very important. In evolutionary terms, the anterior ligament, or anterior talofibular ligament (ATFL), is the youngest and biomechanically the weakest ligament. The middle calcaneofibular ligament (CFL) is the second strongest ligament and also stabilizes the posterior lower ankle (subtalar joint). The posterior talofibular ligament (PTFL) is the strongest.

Injury pattern

The ATFL is the most commonly (58–100%) injured ligament, followed by the CFL combined with the ATFL in 58 to 85% of cases, and the PTFL in only 1.9 to 3%, and then usually only partially (2, 3, 8, 11, 13). An intraoperative analysis of 1235 acute fibular ligament tears during the 1980s, when primary repair was still common (2), showed that the ATFL was torn in 99.6% of cases, of which 21% were isolated, in 74% combined with the CFL, and in 1.9% combined with the CFL and the PTFL. Additional injuries were found in 7.2% and included osteochondral fracture of the talar shoulder (2.4 %), medial malleolar fracture (1.3%), epiphysiolysis of the lateral malleolus (1.0%), rupture of the deltoid ligament (0.8%), the proximal peroneal retinaculum (0.8%), the anterior syndesmotic ligament (0.7%), and the interosseous talocalcaneal ligament (0.2 %).

Nowadays, ultrasound or magnetic resonance imaging (MRI) is used to establish the injury pattern of 96 to 99.5% of cases intended for conservative

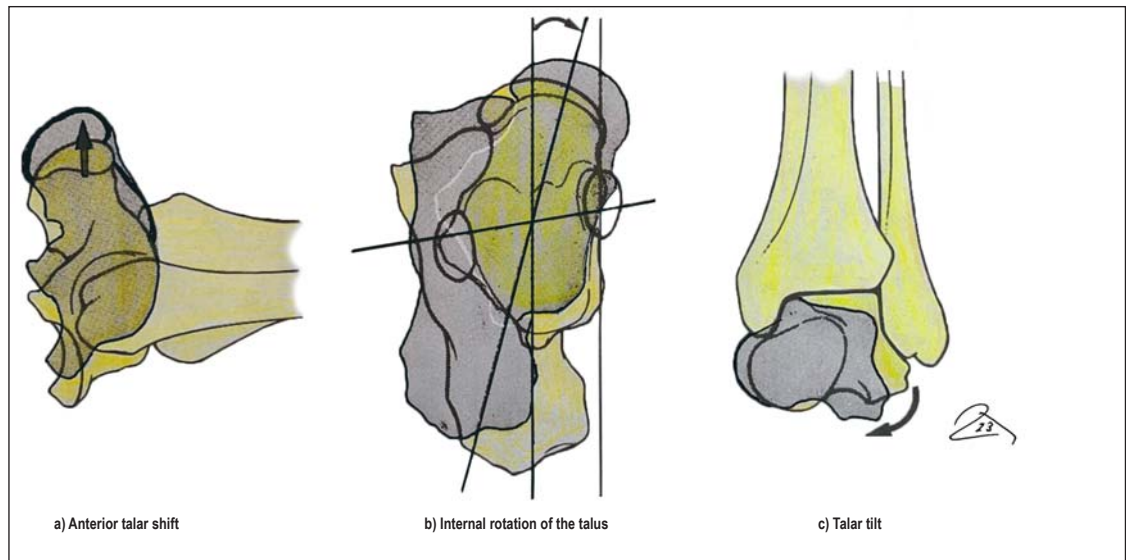


Figure 3: Schematic representation of tear-related anterolateral rotatory instability (ALRI) of the ankle with anterior talar shift, internal rotation, and tilt during clinical tests for stability. a) A pathological talar shift is elicited during the anterior drawer test (ADT) b) anterior talar shift with talar internal rotation during the anterolateral rotatory test c) talar tilt test (TTT), which can only be performed under peripheral nerve block in the presence of an acute tear or after pain has receded (4th to 5th day after rupture) (25)

functional treatment (3, 40). So, in comparison, ultrasound is able to detect tears of the ATFL in 33.3% of cases but those of the CFL in 0%, whereas ten days later, MRI can demonstrate ATFL tears in 50.0% of cases and CFL tears in 6.6% (22). In a comparable number of 104 MRI examinations, Prado et al. (23) found grade 1 to 3 lesions of the ATFL in 100% of cases, of the CFL in 84.8%, and of the PTFL in 46.2%, an additional bone bruise of the talus in 44.2%, and no osteochondral fracture of the talar shoulder.

Diagnostics

The clinical evaluation of tender points (*Figure 2*) is a diagnostic indicator. Of all the nine clinical tests examined in a meta-analysis (24), palpation of the anterior talofibular ligament had the highest sensitivity of 95 to 100%, with a specificity of 0 to 32%.

Ever since the examinations by van Dijk et al. 1996 (25), the greatest importance has been attached to a clinical stability test in the form of the delayed anterior talar drawer test after four to five days when pain is less. It has a sensitivity of 96% and a specificity of 84% in comparison with testing in the first 48 hours with only 71% and 33%, respectively. The clinical talar tilt test is only reliably positive in the acute setting when performed under peripheral nerve block due to the considerable pain associated with it (*Figure 3*) (2). Nowadays, it is only of any clinically relevance when testing for chronic instability.

Apparative diagnostics include radiographs of the ankle in two planes to exclude fracture, with the AP view taken with 20° of internal rotation of the foot to assess exactly the medial clear space for differentiation from syndesmotic injury.

Stress ultrasound has been recommended since 1993 to distinguish an unstable from a stable injury (26). Compared with MRI, which has since found recommendation, a current meta-analysis (27) of ATFL tears demonstrated a higher sensitivity for ultrasound of 88.6% (82.0%–93.5%) compared with MRI of 82.1% (77.1%–86.5%) and a higher specificity of 90.3% (80.1%–96.4%) versus 37.8% (29.1%–47.2%). The meta-analysis also showed that ultrasound detection not only had a high sensitivity of 0.99 and a specificity of 0.92 for an ATFL tear, but also for CFL disruption with 0.95 and 0.99, respectively (28). Because of the additional higher costs associated with MRI, it is only indicated for special cases such as a tear of the PTFL, the syndesmosis, the deltoid ligament, or to detect osteochondral fractures and bruising of the talar cartilage (26–28).

Given that nowadays a distinction between single and double ligament tears with classification of instability has become irrelevant and only the differentiation between stable (stretched ligament) and unstable (tear) is important for initiating conservative functional treatment, hand-held varus stress X-ray views, invasive arthrography of the ankle, and stress tenography of the 1980s have become obsolete. Today, stress X-ray views are recommended for what is called second stage rupture, for recurrent rupture, and to distinguish chronic functional from chronic-mechanical instability (29, 30).

Treatment

Acute phase

Initial treatment of an acute ankle sprain using the RICE protocol (Rest, Ice, Compression, Elevation) can

be enhanced by supplementary multimodal physiotherapy for pain relief, swelling reduction, and return to function (31). According to a multicenter RCT, immobilization in a below-knee cast for ten days is significantly superior to a double-layer tubular compression bandage (Tubigrip), with $p < 0.007$ with respect to pain and ankle function, less significantly for the alternatives Aircast brace and Bledsoe boot (32). Instead of temporary immobilization in a plaster cast, Doherty et al. (33) recommend in their meta-analysis the administration of non-steroidal anti-inflammatory drugs (NSAIDs) combined with early mobilization for analgesia and swelling reduction. In order to avoid potential risks related to impairment of ligament healing from prescribing NSAIDs, Lyrtzis et al. (34) demonstrated that paracetamol in the acute phase is as effective as diclofenac, yet eliminates the risks.

Functional therapy

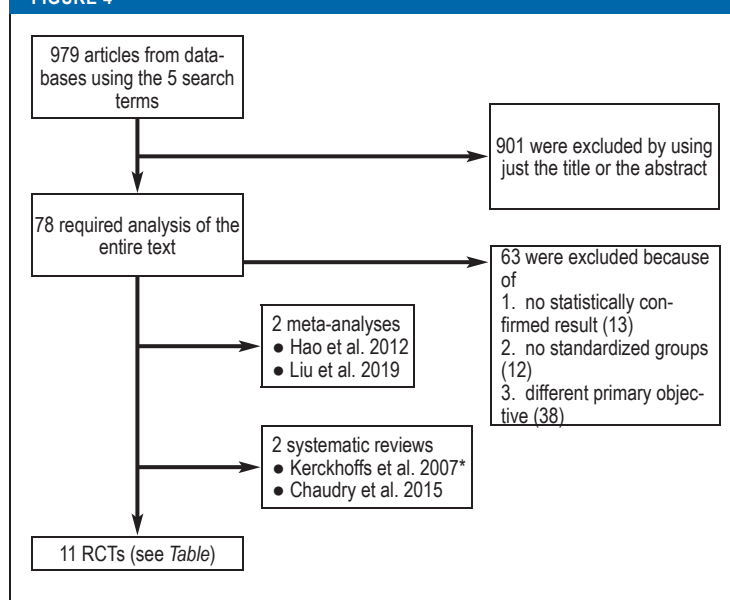
After reduction of swelling and pain, functional therapy is usually initiated with a semi-rigid ankle support for five to six weeks (13, 16, 32, 35) which (evidence based) results in less swelling, a shorter duration of illness, and less skin problems than elastic bandage or tape (35). Other authors prescribe rigid supports (8, 12, 29, 36, 37) for greater mechanical protection and a reliable ligament healing phase. Thus, in experiments, it was shown that only rigid supports highly significantly reduce the tibial-calcaneal angle during a backward drop landing from a 45 cm height onto a 20° slanted surface compared with soft or semirigid supports ($p = 0.003$) and so ensure stable ligament healing (36). Similarly, a study has shown that highly significant protection ($p < 0.0025$) against stress-induced talar tilt and anterior talar shift was demonstrated only for a rigid compared with semirigid support or tape after experimentally induced three-ligament instability (37).

Based on the AWMF (*Association of Scientific Medical Societies in Germany*) S3 guideline on venous thromboembolism prophylaxis, pharmacological prophylaxis is required until mobilization and weight bearing of at least 200 N of the brace-protected ankle (30).

Special therapy

Primary surgical repair followed by functional aftercare is now rare (0.5–4%) and only performed for complete tear of all three ligaments (luxatio supinatoria), tear of all lateral and medial ligaments (luxatio pedis cum talo), additive osteochondral fracture of the talus (Berndt and Harty stage 3 or 4), dislocated avulsion fracture, open tear and/or hemarthros associated with compartment syndrome (3, 30, 40). According to RCTs (29), a more stable joint is achieved after a second stage rupture (tear in an unstable scar) or recurrent rupture (tear in a stable scar) by operative-functional treatment than after conservative-functional therapy ($p < 0.05$), which is why surgery is advised as for chronic-mechanical instability (29, 30).

FIGURE 4



Flowchart of literature search and selection process

* The authors withdrew the article Kerckhoffs et al. 2007 as a systematic review. It is now only available as an intervention review with abstract.
RCT, randomized controlled trial

Aftercare and prevention

In their RCT, Hupperets et al. (38) demonstrated with 522 athletes that a proprioceptive training program using a balance board over eight weeks reduces the risk of recurrent ankle sprain by 35%, even though instructions were only provided in written form. In another RCT, the preventive effect of proprioceptive training was also shown in ice speed skaters, with a p value of 0.001 (39). Given that not only do the collagenous structures tear during each fibular ligament rupture but also the afferent sensory nerve fibers, which are essential for the proprioceptive reflex arc, many authors recommend proprioceptive training not only for prevention in sports but generally for the aftercare of fibular ligament tears and chronic functional instability of the ankle joint (8, 12, 33, 33, 38–40).

Results

The selective analysis (Figure 4) of early prospective randomized studies of the 1980s on surgical versus conservative management of fibular ligament injury (6–11) and the RCTs conducted later around the turn of the millennium until 2017 (13–17) shows that, despite the difference in diagnostic workup (stress X-rays, arthrography, stress tenography), different types of conservative functional treatment (Aircast brace, Caumanns bandage, semirigid and rigid ankle brace), and the use of various scoring systems for follow-up, there are no statistically significant differences in the results of ten of 11 analyzed RCTs (Table). Only Pijnenburg et al. (14) found a better median Powacz score of 26 versus 22 points in their eight-year

TABLE

Synoptic presentation of eleven randomized controlled trials with slightly different approaches, different case numbers of a minimum of n = 25 and a maximum of n = 159 per group, 1/11 with a different level of significance

First author	Surgical	(n)	Conservative	(n)	Outcome	Complications	
						surgical	conservative
Evans 1984 (6)	Suture + plaster cast (3 weeks)	50	Plaster cast (3 weeks)	50	n.s.d.	Deep vein thrombosis + pulmonary embolism (n = 1), sensitive nerve deficit (n = 6), CRPS (n = 1)	DVT (n=1)
Petersen 1985 (7)	Suture + plaster cast (5 weeks)	29	Plaster cast (5 weeks)	30	n.s.d.	not specified	not specified
Zwipp 1986 (8)	Suture + plaster cast (5 weeks) Suture + brace ^{*3} (5 weeks)	50 50	Plaster cast (5 weeks) brace ^{*1} (5 weeks)	50 50	n.s.d.	scar irritation (6/100)	none
Korkalla 1987 (9)	Suture + plaster cast (4–5 weeks)	50	Plaster cast (n = 50)/bandage	50	n.s.d.	DVT (n = 3)	plaster case: DVT (n = 2)
Klein 1988 (10)	Suture + plaster cast (5 weeks)	30	Plaster cast (5 weeks)	30	n.s.d.	not specified	not specified
Möller-Larsen. 1988 (11)	Suture + plaster cast (5 weeks)	55	Plaster cast (n = 55)/tape	55	n.s.d.	not specified	not specified
Povacz 1998 (13)	Suture + plaster cast (6 weeks)	73	Brace ^{*2} (6 weeks)	73	n.s.d.	scar paresthesia (n = 8)	none
Pijnenburg 2003 (14)	Suture + elastic bandage/tape (6 weeks)	159	Elastic bandage/tape (6 weeks)	158	s. d.	excision/skin necrosis (n = 1)	none
Specchiulli 2001 (15)	Suture + plaster cast (5 weeks)	50	Tape (5 weeks)	50	n.s.d.	scar paresthesia (n = 5)	none
Pihlajamäki 2010 (16)	Suture + plaster cast (6 weeks)	25	Brace ^{*2} (6 weeks)	26	n.s.d.	transient scar paresthesia (n = 3)	none
Zhao 2017 (17) ^{*4}	Suture + plaster cast (6 weeks)	60	Plaster cast (6 weeks)	30	n.s.d.	not specified	not specified

CRPS, Complex Regional Pain Syndrome; DVT, deep vein thrombosis; n.s.d., no significant differences; s.d., significant differences for residual pain, recurrent sprains, stability

^{*1} MHH ankle brace; ^{*2} Aircast brace; ^{*3} Tenoplast bandage

^{*4} RCT drafted in Chinese and not available via Medline. The data reproduced here correspond to those of the meta-analysis by Liu F-Q, Diao NC (19)

long-term outcome. With a p value of less than 0.001, they noted significantly more cases of clinically tested positive anterior drawer sign (54% versus 30%, RR 0.64, [0.41; 1.0] in the conservative functional group, even though a positive anterior drawer sign should not be present in any group after stable ligament healing (40). They admitted suboptimal randomization of the study, as 13 cases in the surgical group did not have a torn ligament but only a capsular tear. Follow-up management in both groups comprised neither plaster cast nor brace but a tape dressing or only an elastic bandage in half of the cases in each group, which was a different approach to that of the other RCTs.

With respect to the crucial question of surgical versus conservative management, two meta-analyses (18, 19) of 22 RCTs demonstrated better results for joint stability and higher functional scores in the surgical group, on the one hand, but also complications involving ankylosis, scar tenderness, and sensory loss, on the other, which were not seen in the conservative group. Two Cochrane reviews (20, 21), of which one is only available as an intervention review (20), did not show any statistically significant

superiority of the surgical group with respect to the four primary outcomes (non-return to pre-injury level of sports; ankle sprain recurrence; long-term pain; subjective or functional instability) in 20 RCTs when using the fixed-effect model (20), with the conclusion that there is too little evidence to support any superiority of the surgical group (21).

Discussion

For hardly any other topic in trauma surgery was so much published in the 1970s and 1980s as for the treatment of the acute fibular ligament tear of the ankle. It was not until evidence-based trials were also introduced to trauma surgery that clarity finally emerged in the discussion of this controversial subject.

History of the paradigm shift

Initial prospective randomized trials (6–12) on the treatment of the acute fibular ligament tear covering the surgical versus conservative approach showed that surgery is not superior to conservative and, in particular, conservative-functional management, allowing a therapeutic change to develop during the 1980s. In 1987,

one year after completion of an RCT at Hanover Medical School (8), primary repair was performed there in only 12 of 299 cases (4%) for so-called residual indications (40). Lastly, Swenson et al. (3) analyzed a total of 5373 acute ankle sprains in US high-school athletes between 2005 and 2010 and reported only 0.5% indications for surgery, primarily for severe instabilities with additional rupture of the deltoid ligament in 6.6% of these cases.

Cost savings from the paradigm shift

Apart from the advantage to the patient of not being subjected to the risks of anesthesia and surgery or requiring inpatient care, the paradigm shift also resulted in considerable cost savings for the health care system. For example, calculations conducted in the year 1988 (12) already indicated that an earlier paradigm shift in 1985 alone would have saved 36.8 million deutschmarks for patients insured through the German employer's liability insurance association.

Stability of the paradigm shift even in high-performance sports

Recommendations for high performance athletes should be based on individual characteristics because, according to evidence-based criteria taken from 11 RCTs, two meta-analyses, and two reviews (6–21), conservative and surgical management of a fibular ligament tear appear to be equivalent after weighing up advantages and disadvantages. Furthermore, internationally, no prospective randomized trial of three-ligament tears is available, nor an RCT exclusively involving high-performance athletes. Even Pijnenburg et al. (14), as the only authors of the eleven RCTs (Table) with better eight-year outcomes after surgery, conceded an indication for primary repair only in exceptional cases, in which there is a higher functional demand and when the patient accepts all advantages and disadvantages. Other authors recommend conservative functional management not only for recreational and popular sports, but also for high-performance athletes, emphasizing too the necessity of proprioceptive training during aftercare as well as of preventive measures (39, 40).

Conclusion

Even though Pijnenburg et al. (14) alone differ in their outcomes from the other ten RCTs, they still recommend, like the other authors, the conservative functional approach as best practice. In summary, there are three reasons for this:

- Should conservative treatment fail, the outcome of secondary reconstructions is comparable to primary repair.
- Subjecting all ligament tears to primary surgery would place too much strain on operating room capacity.
- The risks of surgery include infection, abnormal wound healing, nerve injury, and dystrophy, which also result in prolonged periods out of work and

time away from sports as well as higher costs. This is in line with the reasoning of numerous authors (6–13, 15–21, 30, 35, 40).

The fact that the paradigm shift has persisted since the 1980s appears justified in view of the present selective review which includes many documented RCTs, meta-analyses, and reviews after almost 40 years since the first prospective randomized study (6). This is not only of benefit to 96 to 99.5 percent of all patients with acute fibular ligament tears who are spared unnecessary surgery, but it also frees up at least 50 000 surgical slots in hospitals for those cases of unavoidable surgery each year and saves the health care system enormous costs (3, 6–21, 40).

Conflict of interest statement

The author declares that no conflict of interest exists.

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Questions on the article in issue 26/2023:

Conservative Functional Treatment of Acute Fibular Ligament Rupture of the Ankle **cme plus+**

The submission deadline is 29 June 2024. Only one answer is possible per question.
Please select the answer that is most appropriate.

Question 1

Which types of sport are commonly involved in acute fibular ligament injury?

- a) Volleyball, football, basketball
- b) Badminton, swimming, golf
- c) Tennis, inline skating, hockey
- d) Skiing, figure skating, climbing
- e) Cycling, jogging, table tennis

Question 2

According to figures from the German Federal Statistics Office, roughly how many patients were operated upon each year between 2012 and 2021 for a fibular ligament tear?

- a) 500
- b) 1500
- c) 2200
- d) 3000
- e) 5400

Question 3

Which ligament is most commonly torn?

- a) Calcaneofibular ligament
- b) Medial talofibular ligament
- c) Anterior talofibular ligament
- d) Posterior talofibular ligament
- e) Superior calcaneofibular ligament

Question 4

In the text, what does the mnemonic RICE stand for?

- a) "rest, ice, compression, elevation"
- b) "running, ibuprofen, cooling, elevation"
- c) "red, itching, cold, extremity"
- d) "rest, immobilization, cast, exercise"
- e) "running, immunization, cooling, exercise"

Question 5

In which position of the foot is the radiograph of the ankle usually obtained to diagnose a suspected fibular ligament tear?

- a) in 10° of internal rotation of the foot
- b) in 10° of external rotation of the foot
- c) in 5° of internal rotation of the foot
- d) in 15° of external rotation of the foot
- e) in 20° of internal rotation of the foot

Question 6

Which of the tender points palpated for clinical evaluation of the ankle is reported in meta-analyses as having the highest sensitivity?

- a) bifurcate ligament
- b) anterior syndesmotic ligament
- c) lateral calcaneocuboid ligament
- d) anterior talofibular ligament
- e) proximal peroneal retinaculum

Question 7

Which of the following terms refers to disruption of all lateral and medial ligaments?

- a) Luxatio supinatoria
- b) Luxatio pedis cum talo
- c) Luxatio patellae
- d) Luxatio erecta
- e) Luxatio cubiti anterior

Question 8

Which treatment measure was shown in a randomized controlled trial to reduce the risk of a recurrent ankle sprain by 35%?

- a) eight weeks of proprioceptive training with a balance board
- b) wearing an Aircast brace for 14 weeks
- c) stretching exercises six times a day for three weeks
- d) foot hydrotherapy for five weeks
- e) taping for 24 weeks with weight-bearing

Question 9

When was the shift from primary surgical repair to conservative functional treatment?

- a) In the 1960s
- b) In the 1970s
- c) In the 1980s
- d) In the 1990s
- e) In the 2000s

Question 10

Which of the following diagnostic investigations only reaches a sensitivity of 96% four to five days after injury?

- a) Eversion stress test
- b) Ultrasound
- c) Talar tilt test
- d) Talar drawer test
- e) Syndesmosis test